

6.8 均匀平面波斜投射到两理想介质分界面上

入射波: $\vec{E}^i = \vec{E}_0^i e^{-j\vec{k}_i \cdot \vec{r}}$

$$\vec{E}_0^i \perp \hat{k}^i \quad \vec{k}_i = k_1 \hat{k}^i$$

$$\hat{k}^i = \hat{x} \cos \alpha_i + \hat{y} \cos \beta_i + \hat{z} \cos \gamma_i$$

反射波: $\vec{E}^r = \vec{E}_0^r e^{-j\vec{k}_r \cdot \vec{r}}$

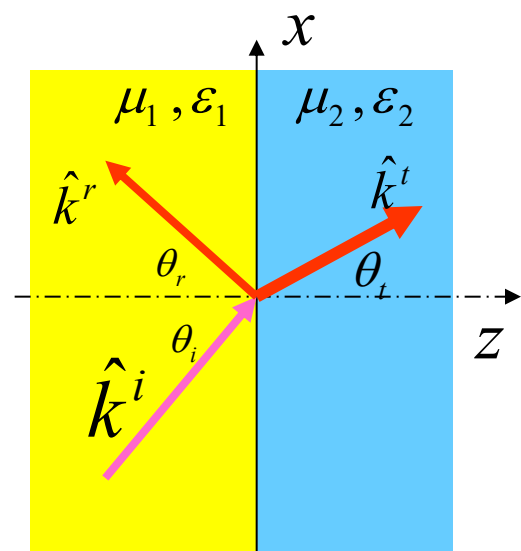
$$\vec{E}_0^r \perp \hat{k}^r \quad \vec{k}_r = k_1 \hat{k}^r$$

$$\hat{k}^r = \hat{x} \cos \alpha_r + \hat{y} \cos \beta_r + \hat{z} \cos \gamma_r$$

透射波: $\vec{E}^t = \vec{E}_0^t e^{-j\vec{k}_t \cdot \vec{r}} \quad \vec{E}_0^t \perp \hat{k}^t$

$$\vec{k}_t = k_2 \hat{k}^t$$

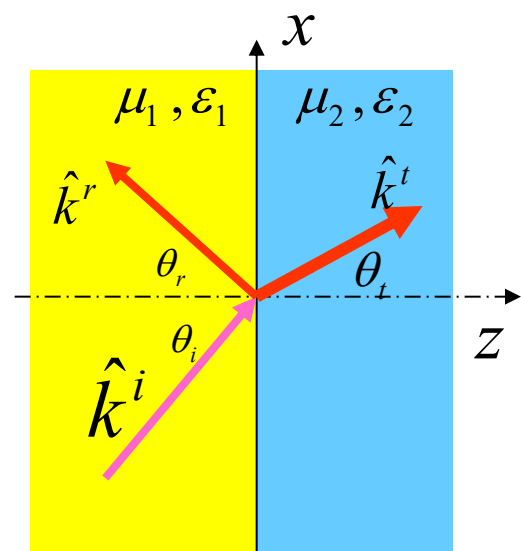
$$\hat{k}^t = \hat{x} \cos \alpha_t + \hat{y} \cos \beta_t + \hat{z} \cos \gamma_t$$



边界条件:

$$E_{1t} = E_{2t}$$

$$\begin{aligned} \hat{z} \times \vec{E}_0^i e^{-j\vec{k}_i \cdot \vec{r}} \Big|_{z=0} + \hat{z} \times \vec{E}_0^r e^{-j\vec{k}_r \cdot \vec{r}} \Big|_{z=0} \\ = \hat{z} \times E_0^t e^{-j\vec{k}_t \cdot \vec{r}} \Big|_{z=0} \end{aligned}$$



(1)传播方向

$$k_1 \underline{x \cos \alpha_i} + k_1 \underline{y \cos \beta_i} + k_1 \cancel{z \cos \gamma_i}$$

$$k_1 \underline{x \cos \alpha_r} + k_1 \underline{y \cos \beta_r} + k_1 \cancel{z \cos \gamma_t}$$

$$k_2 \underline{x \cos \alpha_t} + k_2 \underline{y \cos \beta_t} + k_2 \cancel{z \cos \gamma_t}$$

$$k_1 \cos \alpha_i = k_1 \cos \alpha_r = k_2 \cos \alpha_t \quad \text{if } \beta_i = 90^\circ \quad \beta_r = \beta_t = 90^\circ$$

$$k_1 \cos \beta_i = k_1 \cos \beta_r = k_2 \cos \beta_t \quad \hat{k}^i \quad \hat{k}^r \quad \hat{k}^t \quad \text{在同一平面}$$

$$k_1 \cos \alpha_i = k_1 \cos \alpha_r = k_2 \cos \alpha_t$$

入射角 $\theta_i = \frac{\pi}{2} - \alpha_i$

反射角 $\theta_r = \frac{\pi}{2} - \alpha_r$

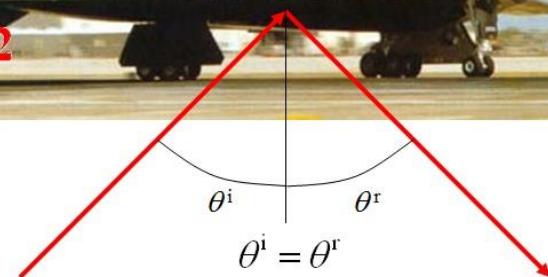
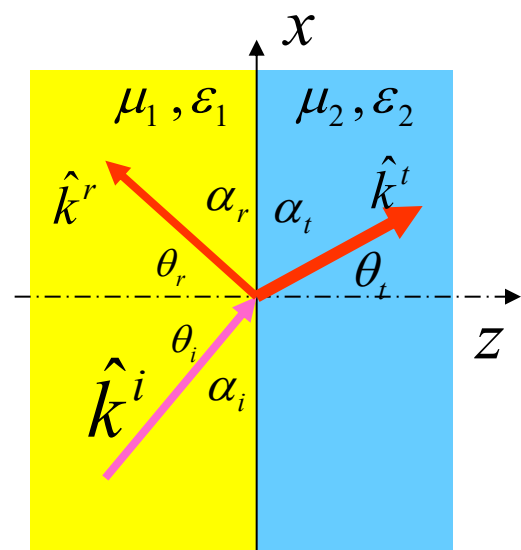
折射角 $\theta_t = \frac{\pi}{2} - \alpha_t$

$$k_1 \sin \theta_i = k_1 \sin \theta_r = k_2 \sin \theta_t$$

$$\theta_r = \theta_i$$

$$\frac{\sin \theta_i}{\sin \theta_t} = \frac{k_2}{k_1} = \frac{\sqrt{\mu_2 \epsilon_2}}{\sqrt{\mu_1 \epsilon_1}} = \frac{c/v_2}{c/v_1} = \frac{n_2}{n_1}$$

Snell Law



$$\vec{E}^i = \vec{E}_0^i e^{-j\vec{k}_i \cdot \vec{r}} \quad \hat{k}^i = \hat{x} \cos \alpha_i + \hat{y} \cos \beta_i + \hat{z} \cos \gamma_i$$

$$\beta_i = 90^\circ \quad \alpha_i = \frac{\pi}{2} - \theta_i \quad \gamma_i = \theta_i$$

$$\begin{aligned} \vec{k}_i \cdot \vec{r} &= k_1 \hat{k}_i \cdot \vec{r} \\ &= k_1 (x \cos \alpha_i + y \cos \beta_i + z \cos \gamma_i) \\ &= k_1 (x \sin \theta_i + z \cos \theta_i) \end{aligned}$$

$$\vec{E}^i = \vec{E}_0^i e^{-jk_1 (x \sin \theta_i + z \cos \theta_i)}$$

$$\vec{E}^r = \vec{E}_0^r e^{-j\vec{k}_r \cdot \vec{r}}$$

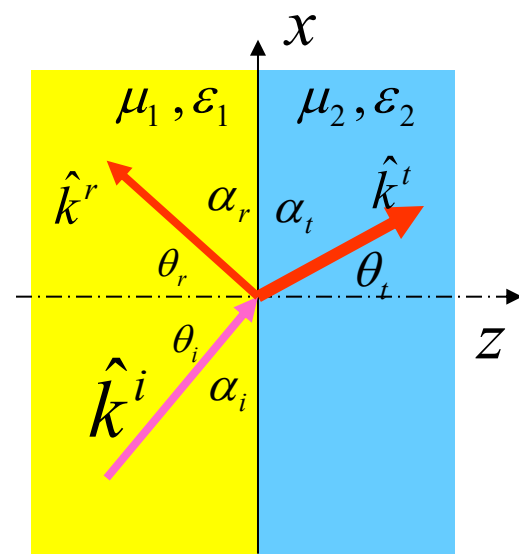
$$\vec{E}^t = \vec{E}_0^t e^{-j\vec{k}_t \cdot \vec{r}}$$

$$\vec{k}_r \cdot \vec{r} = k_1 (x \sin \theta_i - z \cos \theta_i)$$

$$\vec{k}_t \cdot \vec{r} = k_2 (x \sin \theta_t + z \cos \theta_t)$$

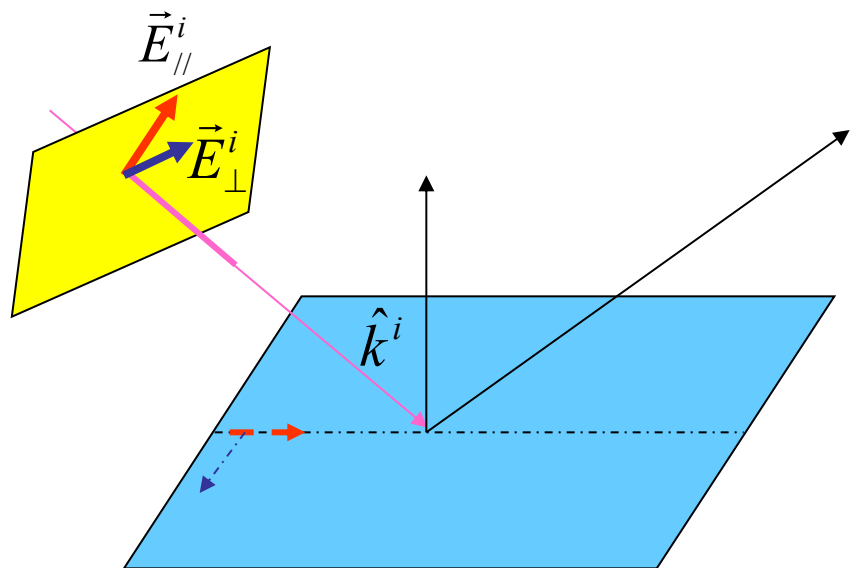
$$\vec{E}^r = \vec{E}_0^r e^{-jk_1 (x \sin \theta_i - z \cos \theta_i)}$$

$$\vec{E}^t = \vec{E}_0^t e^{-jk_2 (x \sin \theta_t + z \cos \theta_t)}$$



(2) 极化方向

$$\vec{E}^i = \vec{E}_{\perp}^i + \vec{E}_{\parallel}^i$$

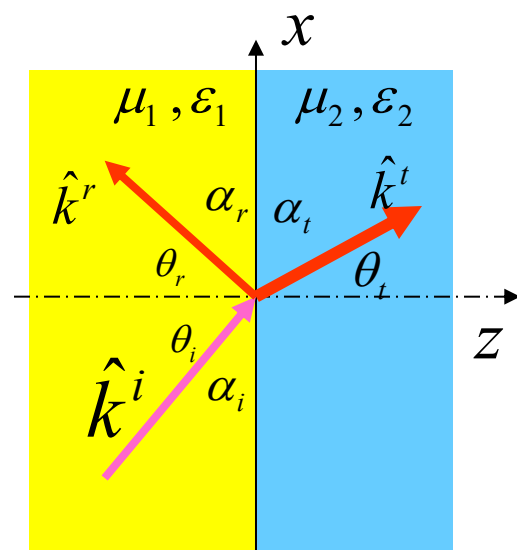


$\vec{E}_{\parallel}^i$ 平行极化波

$\vec{E}_{\perp}^i$ 垂直极化波

$$\vec{E}_{\parallel}^i \rightarrow \vec{E}_{\parallel}^r \quad \vec{E}_{\parallel}^t$$

$$\vec{E}_{\perp}^i \rightarrow \vec{E}_{\perp}^r \quad \vec{E}_{\perp}^t$$



3) 平行极化波

$$\vec{E}_{//}^i = \vec{E}_{0//}^i e^{-jk_1(x \sin \theta_i + z \cos \theta_i)}$$

$$\vec{E}_{//}^r = \vec{E}_{0//}^r e^{-jk_1(x \sin \theta_i - z \cos \theta_i)}$$

$$\vec{E}_{//}^t = \vec{E}_{0//}^t e^{-jk_2(x \sin \theta_t + z \cos \theta_t)}$$

$$\vec{H}_{\perp}^i = \hat{y} \frac{E_{0//}^i}{Z_1} e^{-jk_1(x \sin \theta_i + z \cos \theta_i)}$$

$$\vec{H}_{\perp}^r = \hat{y} \frac{E_{0//}^r}{Z_1} e^{-jk_1(x \sin \theta_i - z \cos \theta_i)}$$

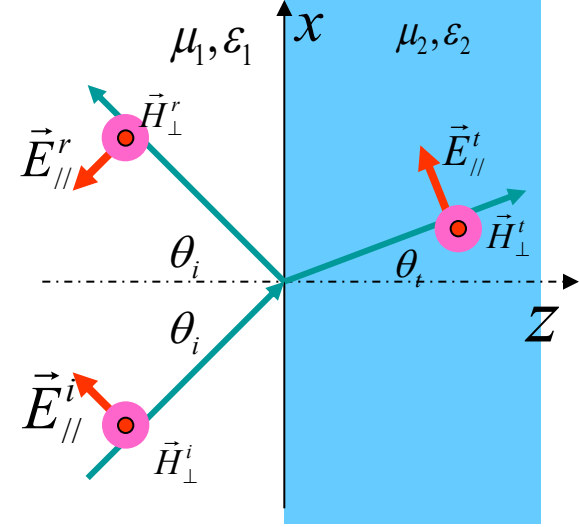
$$\vec{H}_{\perp}^t = \hat{y} \frac{E_{0//}^t}{Z_2} e^{-jk_2(x \sin \theta_t + z \cos \theta_t)}$$

$$\vec{E}_{0//}^i = \underline{E_{0//}^i (\hat{x} \cos \theta_i - \hat{z} \sin \theta_i)}$$

$$\vec{E}_{0//}^r = \underline{E_{0//}^r (-\hat{x} \cos \theta_i - \hat{z} \sin \theta_i)}$$

$$\vec{E}_{0//}^t = \underline{E_{0//}^t (\hat{x} \cos \theta_t - \hat{z} \sin \theta_t)}$$

$$E_{0//}^i \rightarrow E_{0//}^r, E_{0//}^t$$



边界条件

$$E_t : E_{0//}^i \cos \theta_i - E_{0//}^r \cos \theta_i = E_{0//}^t \cos \theta_t \quad H_t : \frac{1}{Z_1} (E_{0//}^i + E_{0//}^r) = \frac{E_{0//}^t}{Z_2}$$

$$R_{//} = \frac{E_{0//}^r}{E_{0//}^i} = -\frac{Z_2 \cos \theta_t - Z_1 \cos \theta_i}{Z_1 \cos \theta_i + Z_2 \cos \theta_t}$$

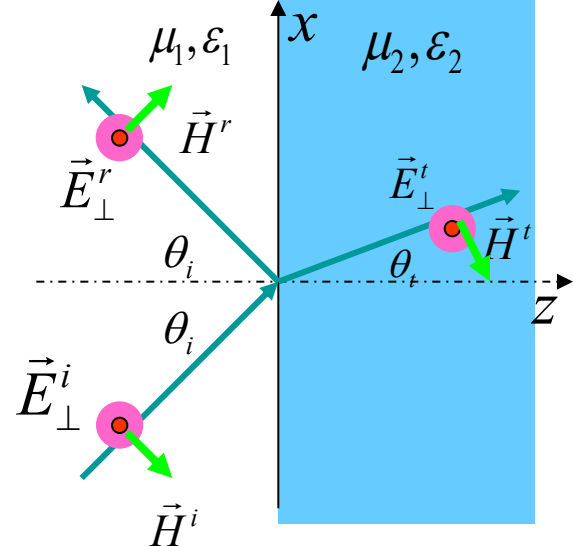
$$T_{//} = \frac{E_{0//}^t}{E_{0//}^i} = \frac{2Z_2 \cos \theta_i}{Z_1 \cos \theta_i + Z_2 \cos \theta_t}$$

4) 垂直极化波

$$\vec{E}_{\perp}^i = \hat{y} E_{0\perp}^i e^{-jk_1(x \sin \theta_i + z \cos \theta_i)}$$

$$\vec{E}_{\perp}^r = \hat{y} E_{0\perp}^r e^{-jk_1(x \sin \theta_i - z \cos \theta_i)}$$

$$\vec{E}_{\perp}^t = \hat{y} E_{0\perp}^t e^{-jk_2(x \sin \theta_t + z \cos \theta_t)}$$



用边界条件可得到:

$$R^{\perp} = \frac{E_{0\perp}^r}{E_{0\perp}^i} = \frac{Z_2 \cos \theta_i - Z_1 \cos \theta_t}{Z_2 \cos \theta_i + Z_1 \cos \theta_t}$$

$$R^{\parallel} = \frac{E_{0\parallel}^r}{E_{0\parallel}^i} = \frac{Z_1 \cos \theta_i - Z_2 \cos \theta_t}{Z_1 \cos \theta_i + Z_2 \cos \theta_t}$$

$$T^{\perp} = \frac{E_{0\perp}^t}{E_{0\perp}^i} = \frac{2Z_2 \cos \theta_i}{Z_2 \cos \theta_i + Z_1 \cos \theta_t}$$

$$T^{\parallel} = \frac{E_{0\parallel}^t}{E_{0\parallel}^i} = \frac{2Z_2 \cos \theta_i}{Z_1 \cos \theta_i + Z_2 \cos \theta_t}$$

R, T 不但与波阻抗有关,还与入射角有关

$$R_{//} = \frac{Z_1 \cos \theta_i - Z_2 \cos \theta_t}{Z_1 \cos \theta_i + Z_2 \cos \theta_t}$$

$$T_{//} = \frac{2Z_2 \cos \theta_i}{Z_1 \cos \theta_i + Z_2 \cos \theta_t}$$

$$R_{\perp} = \frac{Z_2 \cos \theta_i - Z_1 \cos \theta_t}{Z_2 \cos \theta_i + Z_1 \cos \theta_t}$$

$$T_{\perp} = \frac{2Z_2 \cos \theta_i}{Z_2 \cos \theta_i + Z_1 \cos \theta_t}$$

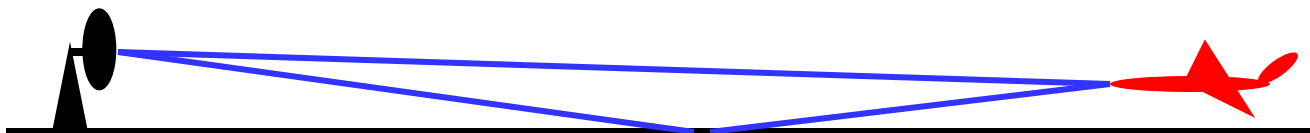
当入射角 $\theta_i \rightarrow 0$ 时，**正投射**：

$$R_{//} = -R_{\perp}$$

当入射角 $\theta_i \rightarrow \frac{\pi}{2}$ 时，**斜滑投射**：

$$R_{//} = R_{\perp} \rightarrow -1 \quad T_{//} = T_{\perp} \rightarrow 0$$

入射波**全**被反射，且反射波与入射波**大小相等**，但**相位相反**。



低空盲区，**地面**雷达无法发现**低空**目标。

4) 无反射和全反射

$$\frac{Z_1}{Z_2} = \sqrt{\frac{\mu_1 / \varepsilon_1}{\mu_2 / \varepsilon_2}} = \sqrt{\frac{\varepsilon_2}{\varepsilon_1}} \quad \sin \theta_t = \frac{k_1}{k_2} \sin \theta_i = \sqrt{\frac{\varepsilon_1}{\varepsilon_2}} \sin \theta_i$$

$$\cos \theta_t = \sqrt{1 - \sin^2 \theta_t} = \sqrt{1 - \frac{\varepsilon_1}{\varepsilon_2} \sin^2 \theta_i}$$

$$R_{//} = \frac{Z_1 \cos \theta_i - Z_2 \cos \theta_t}{Z_1 \cos \theta_i + Z_2 \cos \theta_t} = \frac{\frac{Z_1}{Z_2} \cos \theta_i - \cos \theta_t}{\frac{Z_1}{Z_2} \cos \theta_i + \cos \theta_t}$$

$$= \frac{\sqrt{\frac{\varepsilon_2}{\varepsilon_1}} \cos \theta_i - \sqrt{1 - \frac{\varepsilon_1}{\varepsilon_2} \sin^2 \theta_i}}{\sqrt{\frac{\varepsilon_2}{\varepsilon_1}} \cos \theta_i + \sqrt{1 - \frac{\varepsilon_1}{\varepsilon_2} \sin^2 \theta_i}} = \frac{\frac{\varepsilon_2}{\varepsilon_1} \cos \theta_i - \sqrt{\frac{\varepsilon_2}{\varepsilon_1} - \sin^2 \theta_i}}{\frac{\varepsilon_2}{\varepsilon_1} \cos \theta_i + \sqrt{\frac{\varepsilon_2}{\varepsilon_1} - \sin^2 \theta_i}}$$

同理可得：

$$T^{//} = \frac{2\sqrt{\frac{\epsilon_2}{\epsilon_1}} \cos \theta_i}{\frac{\epsilon_2}{\epsilon_1} \cos \theta_i + \sqrt{\frac{\epsilon_2}{\epsilon_1} - \sin^2 \theta_i}}$$

$$R^{\perp} = \frac{\cos \theta_i - \sqrt{\frac{\epsilon_2}{\epsilon_1} - \sin^2 \theta_i}}{\cos \theta_i + \sqrt{\frac{\epsilon_2}{\epsilon_1} - \sin^2 \theta_i}}$$

$$T^{\perp} = \frac{2 \cos \theta_i}{\cos \theta_i + \sqrt{\frac{\epsilon_2}{\epsilon_1} - \sin^2 \theta_i}}$$

无反射（全折射）：

$$R^{\parallel} = \frac{\frac{\varepsilon_2}{\varepsilon_1} \cos \theta_i - \sqrt{\frac{\varepsilon_2}{\varepsilon_1} - \sin^2 \theta_i}}{\frac{\varepsilon_2}{\varepsilon_1} \cos \theta_i + \sqrt{\frac{\varepsilon_2}{\varepsilon_1} - \sin^2 \theta_i}}$$

无反射条件：

$$\frac{\varepsilon_2}{\varepsilon_1} \cos \theta_i = \sqrt{\frac{\varepsilon_2}{\varepsilon_1} - \sin^2 \theta_i}$$

$$\theta_i = \theta_B = \arcsin \sqrt{\frac{\varepsilon_2}{\varepsilon_1 + \varepsilon_2}}$$

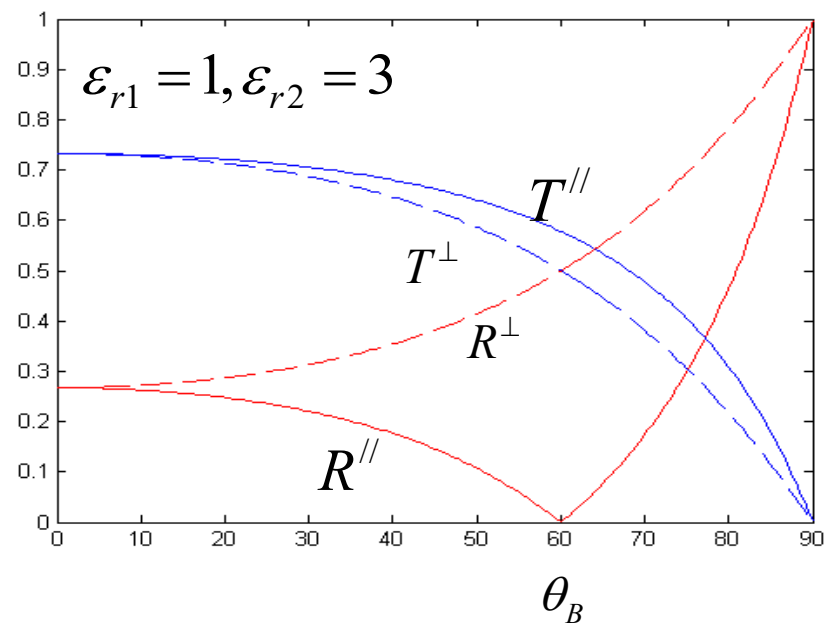
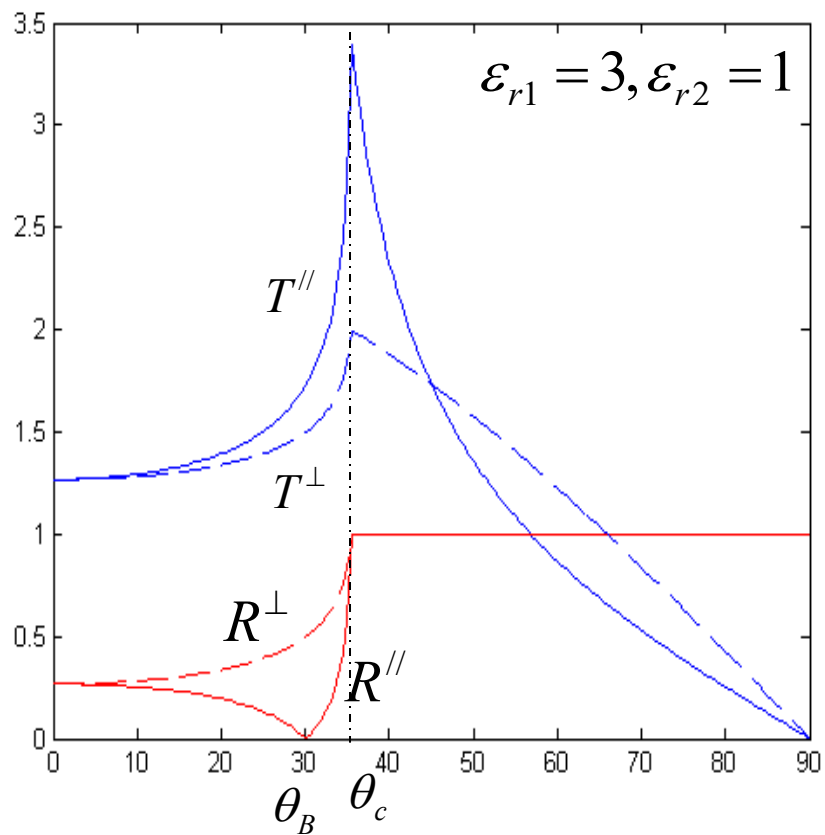
$$R^{\perp} = \frac{\cos \theta_i - \sqrt{\frac{\varepsilon_2}{\varepsilon_1} - \sin^2 \theta_i}}{\cos \theta_i + \sqrt{\frac{\varepsilon_2}{\varepsilon_1} - \sin^2 \theta_i}}$$

$$\cos \theta_i = \sqrt{\frac{\varepsilon_2}{\varepsilon_1} - \sin^2 \theta_i}$$

$$\varepsilon_2 = \varepsilon_1$$

提示：平行极化波会发生无反射现象，对于垂直极化波来说，不可能发生无反射。

$|R|, |T|$ 随入射角的变化曲线



全反射:

$$R^{\parallel} = \frac{\frac{\varepsilon_2}{\varepsilon_1} \cos \theta_i - \sqrt{\frac{\varepsilon_2}{\varepsilon_1} - \sin^2 \theta_i}}{\frac{\varepsilon_2}{\varepsilon_1} \cos \theta_i + \sqrt{\frac{\varepsilon_2}{\varepsilon_1} - \sin^2 \theta_i}}$$

$$R^{\perp} = \frac{\cos \theta_i - \sqrt{\frac{\varepsilon_2}{\varepsilon_1} - \sin^2 \theta_i}}{\cos \theta_i + \sqrt{\frac{\varepsilon_2}{\varepsilon_1} - \sin^2 \theta_i}}$$

$$R^{\parallel} = 1$$

$$\sin^2 \theta_i = \frac{\varepsilon_2}{\varepsilon_1} \longrightarrow$$

$$R^{\perp} = 1$$

$$\longrightarrow \theta_i = \theta_c = \arcsin \sqrt{\frac{\varepsilon_2}{\varepsilon_1}}$$

$$\varepsilon_2 < \varepsilon_1$$

$$\sin \theta_t = \sqrt{\frac{\varepsilon_1}{\varepsilon_2}} \sin \theta_i$$



$$\text{当 } \theta_i = \theta_c \text{ 时, } \theta_t = \frac{\pi}{2}$$

提示：只有当平面波由介电常数较大的光密媒质进入介电常数较小的光疏媒质时，才可能发生全反射现象；而且，当入射角大于临界角时，全反射现象依然存在。

$$\theta_i = \theta_c \quad \theta_t = 90^0$$

$$\sin \theta_t = \sqrt{\frac{\varepsilon_1}{\varepsilon_2}} \sin \theta_i = \begin{cases} = 1 & \theta_i = \theta_c \\ > 1 & \theta_i > \theta_c \end{cases}$$

$$\cos \theta_t = \pm \sqrt{1 - \sin^2 \theta_t} = \pm \sqrt{1 - \frac{\varepsilon_1}{\varepsilon_2} \sin^2 \theta_i}$$

$$\cos \theta_t = \pm j \sqrt{\frac{\varepsilon_1}{\varepsilon_2} \sin^2 \theta_i - 1} \quad \theta_i > \theta_c$$

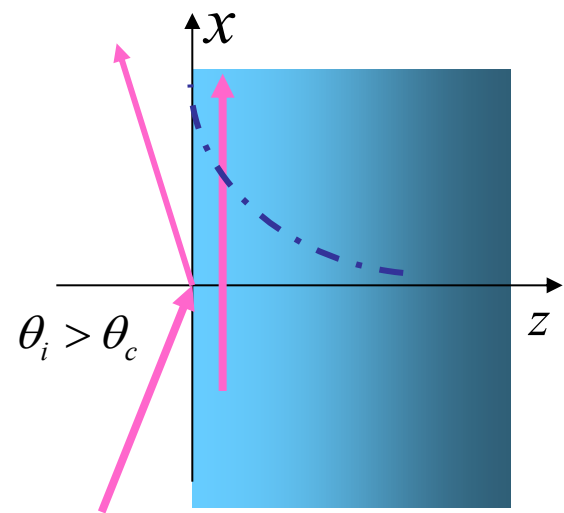
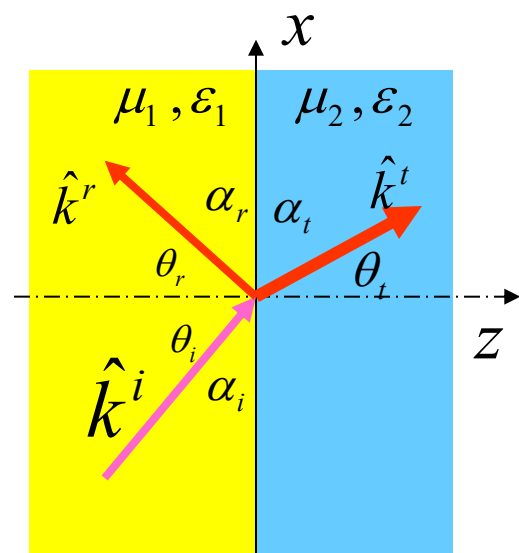
$$\sin \theta_t = \sqrt{\frac{\varepsilon_1}{\varepsilon_2}} \sin \theta_i$$

$$\cos \theta_t = \pm j \sqrt{\frac{\varepsilon_1}{\varepsilon_2} \sin^2 \theta_i - 1}$$

$$\vec{E}^t = \vec{E}_0^t e^{-jk_2(x \sin \theta_t + z \cos \theta_t)}$$

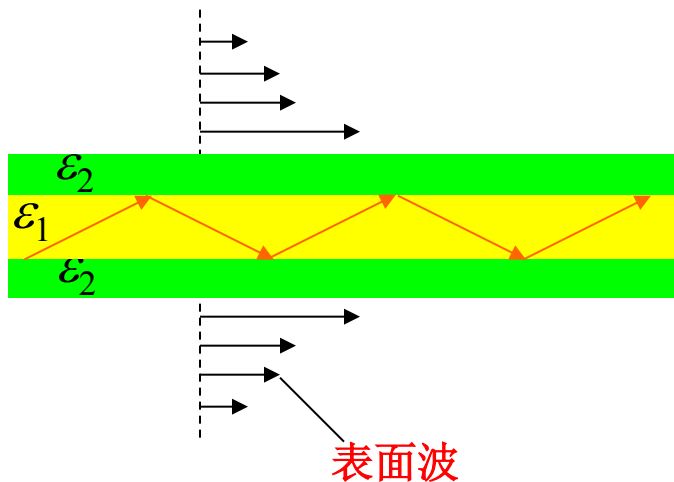
$$= \vec{E}_0^t e^{-jk_2 \sqrt{\frac{\varepsilon_1}{\varepsilon_2}} \sin \theta_i x} e^{-k_2 z \sqrt{\frac{\varepsilon_1}{\varepsilon_2} \sin^2 \theta_i - 1}}$$

提示：折射波沿x方向传播，但其振幅沿正z方向指数衰减，折射波是沿正x方向传播的非均匀平面波，称为表面波



$$E^t = E_0^t e^{-k_2 z \sqrt{(\epsilon_1 / \epsilon_2) \sin^2 \theta_i - 1}} e^{-jk_2 x \sqrt{\epsilon_1 / \epsilon_2} \sin \theta_i}$$

比值 $\frac{\epsilon_1}{\epsilon_2}$ **或入射角** 越大，振幅沿正 z 方向衰减愈快。



一种光导纤维即是由两种介电常数不同的介质层形成的，其芯线介电常数**大于**外层介电常数。

当光束以**大于**临界角的入射角度自芯线内部向边界投射时，即可发生**全反射**，光波局限在芯线内部传播，这就是光导纤维的**导波**原理。

6.9 均匀平面波斜投射到理想导电平面上

1) 平行极化

$$\vec{E}^i = E_0^i (\hat{x} \cos \theta - \hat{z} \sin \theta) e^{-jk(x \sin \theta + z \cos \theta)}$$

$$\vec{E}^r = E_0^r (-\hat{x} \cos \theta - \hat{z} \sin \theta) e^{-jk(x \sin \theta - z \cos \theta)}$$

$$\vec{H}^i = \hat{y} \frac{E_0^i}{Z} e^{-jk(x \sin \theta + z \cos \theta)}$$

$$\vec{H}^r = \hat{y} \frac{E_0^r}{Z} e^{-jk(x \sin \theta - z \cos \theta)}$$

边界条件 $E_t = 0 \quad E_0^i \cos \theta - E_0^r \cos \theta = 0 \quad E_0^r = E_0^i$

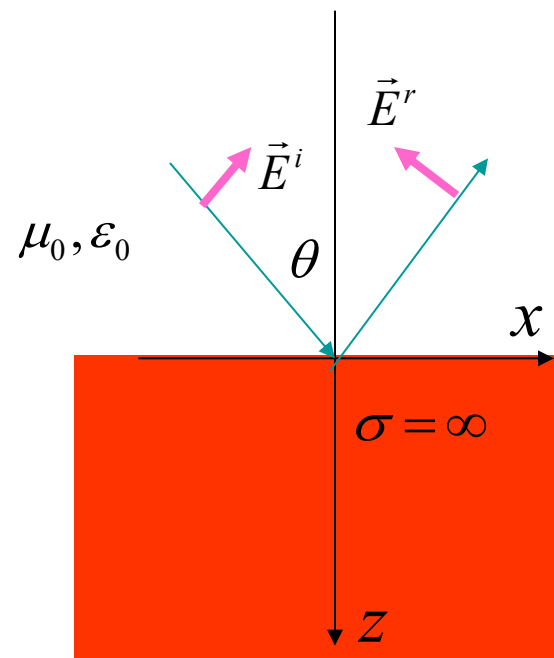
$$\vec{E} = \vec{E}^i + \vec{E}^r \quad E_x = E_0^i \cos \theta e^{-jk(x \sin \theta + z \cos \theta)} - E_0^i \cos \theta e^{-jk(x \sin \theta - z \cos \theta)}$$

$$= E_0^i \cos \theta (e^{-jkz \cos \theta} - e^{jkz \cos \theta}) e^{-jkx \sin \theta}$$

$$E_x = -j2E_0^i \cos \theta \sin(kz \cos \theta) e^{-jkx \sin \theta}$$

$$E_z = -2E_0^i \sin \theta \cos(kz \cos \theta) e^{-jkx \sin \theta}$$

$$\vec{H} = \vec{H}^i + \vec{H}^r \quad H_y = 2 \frac{E_0^i}{Z} \cos(kz \cos \theta) e^{-jkx \sin \theta}$$



向x方向传播的平面波

非均匀平面波

非 TEM 波 TM

$$v'_p = \frac{\omega}{k \sin \theta} = \frac{c}{\sin \theta} > c$$

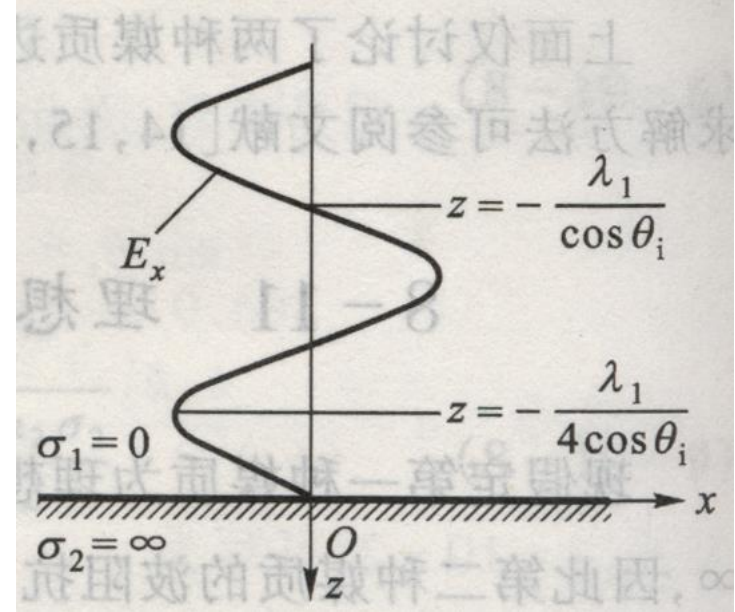
$$E_x = -j2E_0^i \cos \theta \sin(kz \cos \theta) e^{-jkx \sin \theta}$$

$$E_z = -2E_0^i \sin \theta \cos(kz \cos \theta) e^{-jkx \sin \theta}$$

$$H_y = 2 \frac{E_0^i}{Z} \cos(kz \cos \theta) e^{-jkx \sin \theta}$$

$$\vec{S}_c = \vec{E} \times \vec{H}^* = (\hat{x}E_x + \hat{z}E_z) \times \hat{y}H_y^*$$

$$\vec{S}_c = \hat{x}4 \frac{(E_0^i)^2}{Z} \sin \theta \cos^2(kz \cos \theta) - j\hat{z}4 \frac{(E_0^i)^2}{Z} \cos \theta \sin(kz \cos \theta) \cos(kz \cos \theta)$$



在x方向上存在单向的能量流动，在z方向上只有电磁能量的相互交换。

2) 垂直极化

$$\vec{E}^i = \hat{y} E_0^i e^{-jk(x \sin \theta + z \cos \theta)}$$

$$\vec{H}^i = \frac{E_0^i}{Z} (-\hat{x} \cos \theta + \hat{z} \sin \theta) e^{-jk(x \sin \theta + z \cos \theta)}$$

$$\vec{E}^r = \hat{y} E_0^r e^{-jk(x \sin \theta - z \cos \theta)}$$

$$\vec{H}^r = \frac{E_0^r}{Z} (\hat{x} \cos \theta + \hat{z} \sin \theta) e^{-jk(x \sin \theta - z \cos \theta)}$$

$$E_y = -j2E_0^i \sin(kz \cos \theta) e^{-jkx \sin \theta}$$

$$H_x = -2 \frac{E_0^i}{Z} \cos \theta \cos(kz \cos \theta) e^{-jkx \sin \theta}$$

$$H_z = -j2 \frac{E_0^i}{Z} \sin \theta \sin(kz \cos \theta) e^{-jkx \sin \theta}$$

向x方向传播的平面波 非均匀平面波

非 *TEM* 波

TE

